

ECSE 2000

Lec 7

Feb 2

Topic: Conditional Probability (cont)

Notion of independent events

□ Example on Spam email Filtering

We get tons of emails especially Russian spam emails.

So I will train my spam filter by putting Russian Spam to my junk mailbox. The spam filter compute the conditional probability

$$P(\text{Russian} | \text{spam})$$

It also knows $P(\text{spam})$ and $P(\text{Russian})$ by counting my mail/junk boxes. Then once we have a good estimate of the above three probabilities, we are interested in

$$P(\text{Spam} | \text{Russian}) = \frac{P(\text{Russian} | \text{spam}) P(\text{Spam})}{P(\text{Russian})}$$

Remark: If the above probability is high ($\geq 80\%$),
apply a rule to automatically delete emails.

□ Independent Events

In high level, Events A and B are independent
if we know that Event A occurs has nothing to
do with the probability of Event B.

In previous example,

Rolling two 6-sided Dice,

$$\begin{aligned} P(\text{Die 1} = 2, \text{Die 2} = 3) &= P(\text{Die 1} = 2) \cdot P(\text{Die 2} = 3) \\ &= \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36} \end{aligned}$$

This matches the definition of independent events:
formally, A and B are independent when

$$P(A \cap B) = P(A) \cdot P(B).$$

Based on this definition, we can also have (if $P(B) \neq 0$)

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A) \cdot P(B)}{P(B)} = P(A).$$

That is, the conditional probability of A given B is the same as $P(A)$ - meaning that knowing Event B doesn't have any effect on how likely A is.

Note

Independence is not same as **mutually exclusive**
i.e., $A \cap B = \emptyset$, which means that A and B
cannot happen at the same time.

To explain this point, consider the following example:

Ex. Roll a fair die. Define two events.

$$A = \{4\}, \quad B = \{2, 4, 6\}.$$

$$P(A) = \frac{1}{6}, \quad P(B) = \frac{3}{6} = \frac{1}{2}$$

$$P(A \cap B) = P(\{4\}) = \frac{1}{6}$$

$$\neq P(A) \cdot P(B) = \frac{1}{12}$$

→ A and B not independent

$$P(A|B) = \frac{1}{3} \neq P(A)$$

$$P(B|A) = 1 \neq P(B)$$

Property In fact, if $A \subset B$ and

$$0 < P(A) \leq P(B) < 1$$

then Events A and B will be never independent.

$$P(B|A) = 1 \neq P(B) < 1$$

Note Events can be independent while still seeming "related."

Example: Roll a die. Define two events

$$A = \{3, 4\}, \quad B = \{2, 4, 6\}.$$

$$P(A) = \frac{1}{3}, \quad P(B) = \frac{1}{2}$$

$$P(A \cap B) = P(\{4\}) = \frac{1}{6}$$

$$= P(A) \cdot P(B) = \frac{1}{2} \times \frac{1}{3}$$

✓ Yes, they are independent.

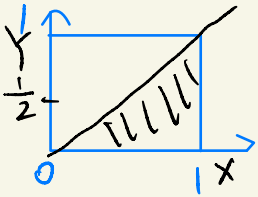
Remark: So knowing A happened doesn't change probability of Event B , even though they have outcomes in common. This is the place that seems counter-intuitive.

The only way to be sure is to check the definition of independence

$$P(A \cap B) = P(A) P(B).$$

□ Similar ideas with continuous sample spaces.

Pick X and Y uniformly at random from $[0, 1]$.



$$A = \{X > Y\}, \quad P(A) = \frac{1}{2}$$

$$B = \{Y > \frac{1}{2}\}, \quad P(B) = \frac{1}{2}$$

$$C = \{X > \frac{1}{2}\}, \quad P(C) = \frac{1}{2}$$

$$P(A \cap B) = P(X > Y, Y > \frac{1}{2})$$

$$= \frac{1}{8} \neq P(A) \cdot P(B)$$

$$P(A \cap C) = \frac{3}{8} \neq P(A) \cdot P(C)$$

$$P(B \cap C) = \frac{1}{4} = P(B) \cdot P(C)$$

Remark: B and C are independent, while A is not independent with others.

In general, for a set of events $\{A_1, A_2, \dots, A_k\}$.
To be independent, we will have

pairwise $P(A_i A_j) = P(A_i) P(A_j), \forall i \neq j$

Triples $P(A_i A_j A_k) = P(A_i) P(A_j) P(A_k),$
 $\vdots$
 $\forall i \neq j \neq k$

k events $P\left(\bigcap_{i=1}^k A_i\right) = \prod_{i=1}^k P(A_i)$

Example: Roll 4-sided die. Define three events

$$A = \{1, 2\}, B = \{1, 3\}, C = \{1, 4\}$$

$$P(A) = P(B) = P(C) = \frac{1}{2}.$$

$$P(A \cap B) = P(\{1\}) = \frac{1}{4}$$

$$P(B \cap C) = P(\{1\}) = \frac{1}{4}$$

$$P(A \cap C) = P(\{1\}) = \frac{1}{4}$$

$$P(A \cap B \cap C) = P(\{1\}) = \frac{1}{4} \neq P(A)P(B)P(C)$$

They are not independent!

□ Many real systems can be viewed as sequence of independent experiments.

i.e., Experiments $E_1, E_2, \dots, E_n$

Outcomes $(s_1, s_2, \dots, s_n)$

Sample space $s_1 \times s_2 \times \dots \times s_n$

Special case: Bernoulli Trial

Each sample space s_i is binary

The example of flipping the coins multiple times is an example of Bernoulli Trial.

$$S_i = \{ \text{Head, Tail} \}$$

$$= \{ \text{Success, Fail} \}$$

Assume $P(\text{Success}) = p$, $P(\text{Failure}) = 1-p$.

Example: Flip a coin 5 times.

What is $P(\text{HTHTH})$
sequence of outcomes

Since each coin flip is independent,

$$\begin{aligned} P(\text{HTHTH}) &= P(H) P(T) P(H) P(T) P(H) \\ &= p^3 (1-p)^2 \end{aligned}$$

Generalize this, we can calculate

$$P(\text{any sequence}) = p^{\# \text{ of heads}} (1-p)^{\# \text{ of tails}}$$

What is

$$P(3 \text{ out of } 5 \text{ heads}) = C_3^5 P^3 (1-p)^2$$

Generalize

$$\Rightarrow P(k \text{ out of } n \text{ flips heads}) = C_k^n P^k (1-p)^{n-k},$$

$k=1, 2, \dots, n$

This is called the Bernoulli Probability Law.